Stat 451: Group 12 Project Report

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Introduction

We analyzed the Spotify Tracks Dataset to identify which audio features predict song popularity and how well it can be modeled. Our primary question was: how well can we predict song popularity from audio features? Key variables included danceability, energy, loudness, tempo, and other audio characteristics.

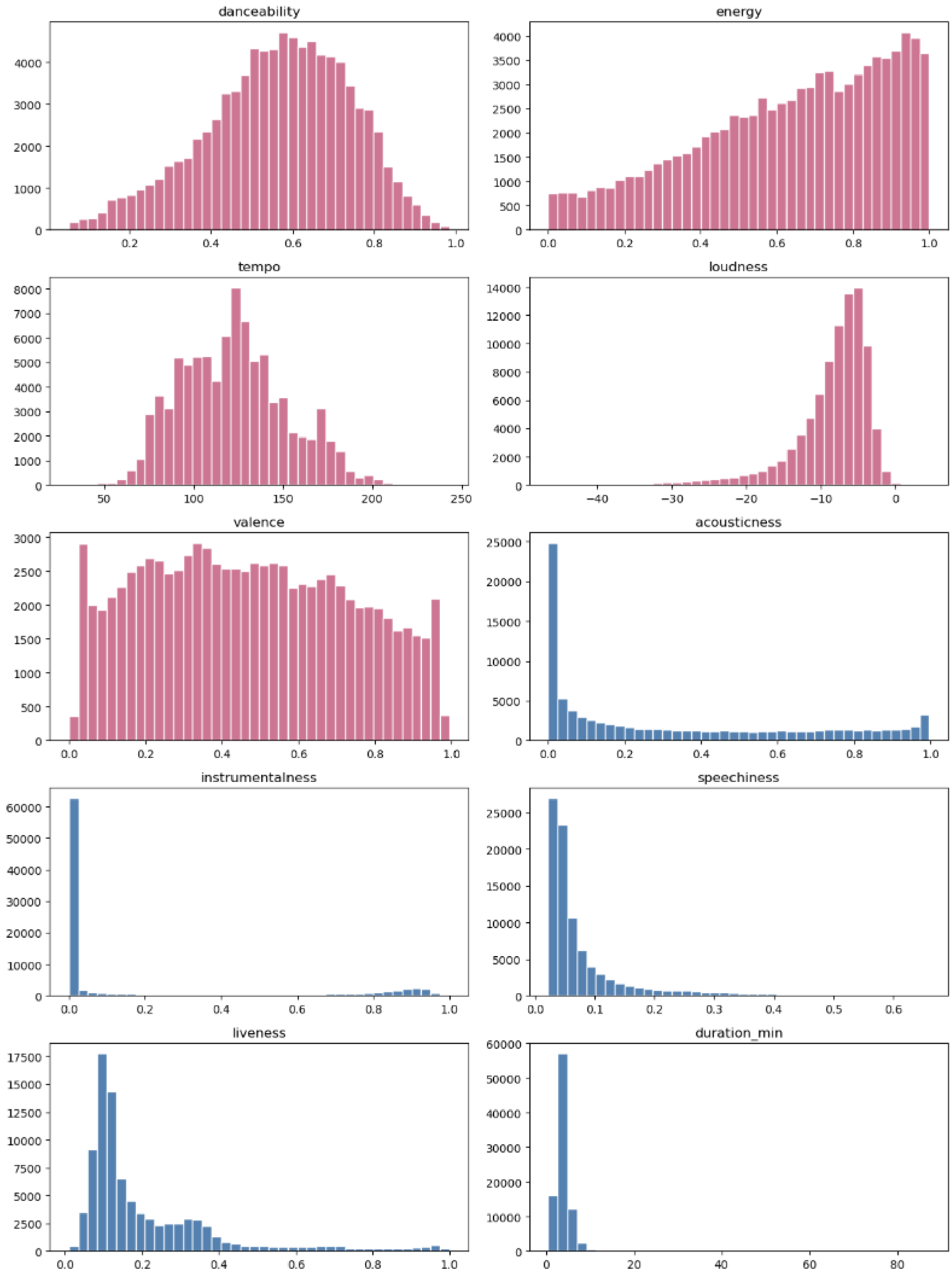
We conducted exploratory data analysis, built predictive models, and evaluated them using appropriate metrics. Results showed that danceability and loudness are linked to higher popularity, while instrumental and speech-heavy tracks are less popular. Final model performance was moderate, indicating audio features explain only part of popularity, with external factors also playing a role.

Data Source

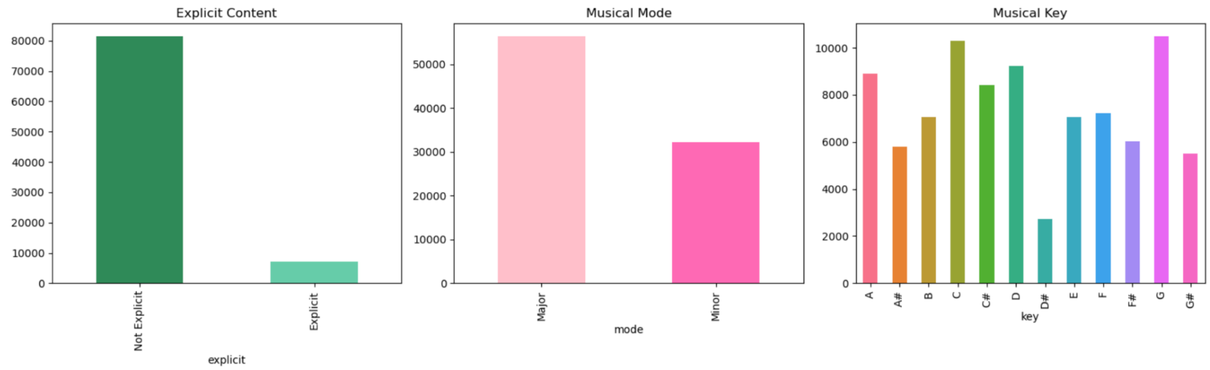
The Spotify Tracks Dataset was sourced from Spotify's Web API and published on Kaggle. It originally contained 114,000 rows, 21 columns, and 125 genres. Features used in our analysis include danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo, duration, and genre.

The data was cleaned by removing missing values and duplicates, filtering non-music tracks (speechiness > 0.66), removing tracks shorter than 30 seconds or with 0 tempo, encoding categorical variables, and converting duration to minutes. After cleaning, the dataset contained 88,704 observations. For modeling, track_genre was one-hot encoded, so genre could be included as a feature, increasing the number of predictors as well.

Distribution of Numerical Audio Features:



Distribution of Categorical Audio Features:



Methods

We built predictive models using OLS, Ridge, and Lasso regression, along with decision trees. The dataset was split into training, validation, and test sets (80/10/10), with linear models standardized while the decision tree used raw values. Models were tuned using 5-fold cross-validation on the training set via GridSearchCV with R^2 as the scoring metric. Performance was evaluated using R^2, MSE, and RMSE.

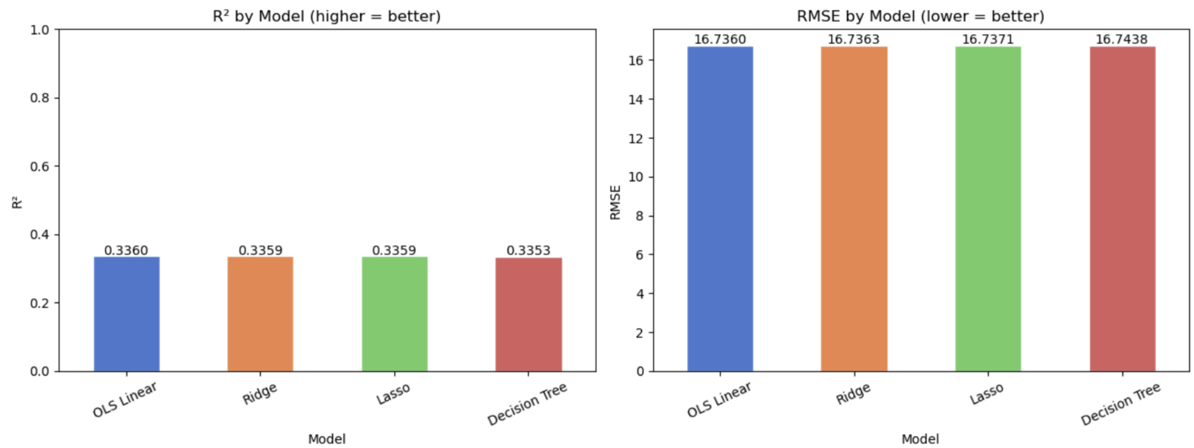
Results

All models performed similarly ($R^2 \approx 0.33$, RMSE ≈ 16.7), indicating that increasing model complexity did not meaningfully improve predictive accuracy. Including genre improved model performance compared to using audio features alone, highlighting its importance in predicting popularity. Final OLS results on the unseen test set ($R^2 \approx 0.35$, RMSE ≈ 16.5) were consistent with validation results, confirming strong generalization to unseen data and not overfitting.

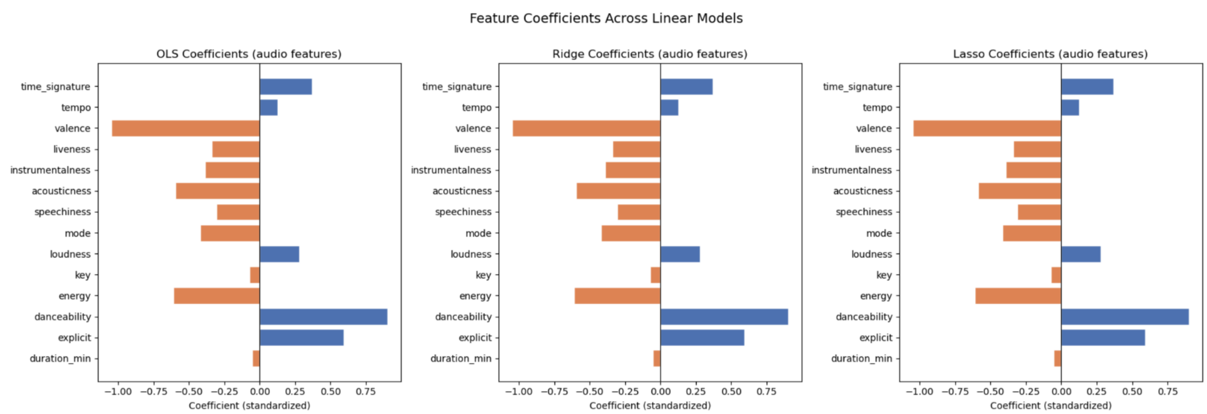
Across models, track genre was the strongest overall predictor, while the most influential audio features included danceability, valence, acousticness, and energy. Overall, these results suggest while genre and audio features are meaningful predictors, they only explain about one-third of the variation in popularity.

Regression Analysis Including Genre:

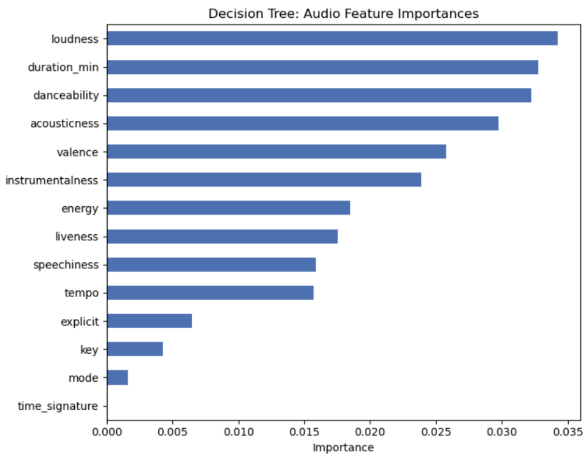
Model	R^2	MSE	RMSE
OLS Linear	0.3360	280.0951	16.7360
Ridge	0.3359	280.1030	16.7363
Lasso	0.3359	280.1291	16.7371
Decision Tree	0.3353	280.3545	16.7438

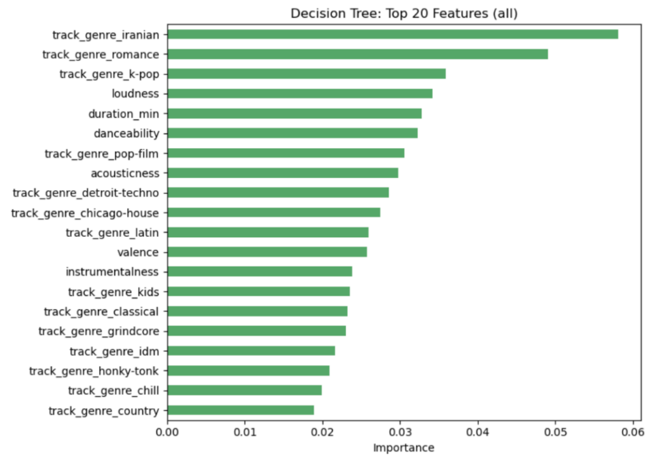


Feature Coefficients Across Linear Models (excluding genre):



Decision Tree of Audio Feature Importance & Top Features:





Conclusion

Audio features and genre provide moderate predictive power for song popularity, explaining approximately 33% of its variation. This suggests that non-audio factors such as marketing, artist reputation, and cultural trends likely play a larger role in determining popularity.

One limitation of this study is the restricted feature set, which excludes external influences on popularity. Additionally, the dataset is derived from Spotify, which may introduce bias toward certain regions or genres. Future work could incorporate external data sources and compare results across platforms such as Apple Music.

Contributions

Member	**Proposal**	**Coding**	**Presentation**	**Report**
Arianna Barbosa	1	1	1	1
Greta Fogel	1	1	1	1
Lexi Rush	1	1	1	1
Levi Hellenbrand	1	1	1	1
Eirfann Danish	1	1	1	1
Mukhriz Hazli	1	1	1	1